StyleSlider: Controllable Geometric Stylization of 3D Meshes

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1. Summary

Recent diffusion-guided 3D stylization methods can produce expressive geometric transformations, including exaggerated proportions, elongated features, and cartoon-like shapes. However, controlling the exact amount of geometric deformation remains challenging.

We propose **StyleSlider**, an interactive system that allows users to continuously adjust the strength of geometric stylization on a 3D mesh. By interpolating vertex deformation fields or cage-based deformation parameters, our system will generate intermediate stylizations between the original mesh and a fully stylized target. We will investigate how to make these transitions visually smooth, predictable, and geometrically valid.

2. Problem Description

Geometric stylization is an important technique in computer graphics, particularly for animation, character design, digital sculpting, and artistic 3D modeling. Unlike traditional style transfer methods that primarily modify colors or textures, geometric stylization changes the actual shape of a 3D object.

Recent methods such as GeoStyle demonstrate that diffusion models can guide substantial geometric changes while preserving the original object's semantic identity. For example, a character's limbs can become elongated, its head can become exaggerated, or its overall silhouette can be modified to match a reference artistic style.

However, diffusion-guided deformation typically involves an optimization process that produces a stylized result without necessarily providing direct, predictable control over the amount of shape exaggeration. Users may prefer a moderate transformation rather than the strongest stylized output. Re-running the optimization with different parameters can also be expensive and may yield inconsistent results.

Our project addresses the following question:

Can we create a continuous and intuitive control over geometric stylization strength while maintaining smooth and valid mesh deformation?

We propose adding an explicit stylization parameter, $\alpha \in [0, 1]$, where:

- $\alpha = 0$ represents the original mesh.
- $\alpha = 0.5$ represents an intermediate stylization.
- $\alpha = 1$ represents the fully stylized target mesh.

Our main approach is to interpolate between the original mesh and its stylized deformation, rather than repeatedly running a diffusion model.

For an original mesh with vertices V_0 and a corresponding stylized mesh with vertices V_1, a basic interpolation is:

$$V(\alpha) = V_0 + \alpha(V_1 - V_0)$$

We will also investigate interpolation of cage control points. Because cage deformation operates on a lower-dimensional representation, it may produce smoother and more coherent geometric changes than direct vertex interpolation.

A major technical challenge is that interpolation does not automatically guarantee high-quality geometry. Intermediate meshes may exhibit unwanted stretching, self-intersections, collapsed triangles, or visually uneven deformation. We therefore aim to evaluate the relationship between stylization strength, geometric quality, and user-perceived exaggeration.

The focus of the project is geometric deformation and interactive graphics rather than training a new generative model.

3. Goals and Deliverables

3.1 What We Plan to Deliver (Baseline)

Goal 1: Implement controllable mesh stylization.

We will develop a geometric deformation pipeline that takes an original 3D mesh and a corresponding stylized target mesh as input.

The system will allow continuous interpolation between the two shapes using a user-defined parameter α. We will first implement linear interpolation of vertex displacement fields. Since the original and stylized meshes are expected to have matching topology and vertex correspondence, each vertex can be tracked throughout the deformation.

We will test the system using at least three 3D objects with visibly different geometric characteristics, such as an animal, a humanoid character, and a simple decorative object.

Expected output: For each object, we will render a sequence of meshes at five stylization strengths: 0%, 25%, 50%, 75%, and 100%.

Goal 2: Implement cage-based deformation control.

We will explore a second approach using a coarse deformation cage surrounding the original mesh.

Instead of directly interpolating all mesh vertices, we will interpolate the positions of cage control points and propagate the resulting deformation to the enclosed mesh.

We will compare vertex interpolation and cage-based interpolation in terms of deformation smoothness, geometric distortion, and computational performance.

Expected output: Side-by-side comparisons of the two interpolation methods, highlighting differences in silhouette, mesh deformation, and geometric quality.

Goal 3: Create an interactive 3D visualization system.

We will develop an interactive demo that allows users to load a mesh and adjust its stylization strength through a slider.

The interface will provide:

- A slider for adjusting stylization strength from 0% to 100%.
- An interactive 3D viewer with camera rotation and zoom.
- Real-time updates to the deformed mesh.
- The ability to switch between multiple example meshes.
- A comparison between the original and current stylized geometry.

Our goal is to make the interpolation sufficiently efficient for smooth interactive manipulation.

Expected output: A live demonstration and a short video showing geometric exaggeration changing continuously as the slider moves.

Goal 4: Evaluate geometric quality and controllability.

We will conduct quantitative and qualitative evaluations to understand whether a simple stylization-strength parameter provides meaningful control.

We will measure:

1. **Geometric validity:** The number of degenerate triangles, inverted or flipped faces, and detected self-intersections across different strength values.
2. **Deformation smoothness:** Changes in neighboring vertex displacements and local edge-length distortion. We will examine whether interpolation creates abrupt or uneven changes.
3. **Monotonic stylization strength:** We will measure displacement from the original mesh and investigate whether greater slider values consistently correspond to greater

geometric changes. We will distinguish this from perceptual exaggeration, which may not follow a linear relationship.

4. **Performance:** We will measure the time required to update the mesh at different strength values and compare vertex-based and cage-based interpolation.

We will generate graphs showing these metrics as functions of the stylization parameter.

Expected output: A quantitative evaluation containing displacement-versus-strength plots, geometric-distortion comparisons, and performance measurements.

3.2 What We Hope to Deliver (Aspirational)

Stretch Goal 1: Geometry-aware interpolation.

Linear interpolation is computationally simple but may produce undesirable intermediate shapes. If time permits, we will investigate geometry-aware interpolation using Laplacian smoothing or As-Rigid-As-Possible (ARAP) regularization.

We will compare these techniques against simple linear interpolation to determine whether additional geometric constraints improve intermediate results.

Stretch Goal 2: Perceptually uniform stylization control.

A linear interpolation parameter does not necessarily correspond to a linear increase in perceived stylization.

For example, changing a character's head size from 100% to 120% may be much less noticeable than changing it from 140% to 160%.

We hope to investigate whether a nonlinear mapping from slider value to deformation strength can produce more perceptually uniform increments.

If feasible, we will perform a small user study in which participants compare intermediate shapes and rate their perceived stylization strength.

Stretch Goal 3: Localized stylization-strength control.

We hope to extend the system to support different stylization strengths in different regions of a mesh.

For example, users could exaggerate a character's head while keeping its body nearly unchanged, or elongate an animal's legs without modifying its torso.

This extension would require blending local deformation fields while maintaining smooth transitions between edited and unedited regions.

4. Planned Results and Evaluation

Our main visual result will be a comparison of the same object at multiple stylization strengths.

For example, we plan to use a stylized animal model and generate the following sequence:

Strength	Expected Visual Result
0%	Original 3D mesh
25%	Subtle changes to proportions
50%	Moderate geometric exaggeration
75%	Stronger stylization while preserving recognizability
100%	Fully stylized target mesh

This sequence will be rendered from consistent camera viewpoints and lighting conditions so that changes can be compared directly.

We will also produce side-by-side visualizations of linear vertex interpolation and cage-based deformation, together with wireframe overlays that reveal geometric stretching or distortions.

Our quantitative evaluation will include plots of geometric displacement, local distortion, and mesh update time against stylization strength.

We aim to answer three main research questions:

1. **Controllability:** Does interpolation provide a smooth and predictable progression from the original mesh to the stylized target?
2. **Geometric quality:** Which interpolation strategy better preserves mesh quality, particularly for large geometric exaggerations?
3. **Efficiency:** Can stylization strength be adjusted interactively without rerunning expensive generative optimization?

We will use fixed original–target mesh pairs for comparisons so that the deformation methods can be evaluated under identical conditions.

5. Implementation Plan

Our implementation will consist of three components.

Mesh processing. We will load original and stylized meshes, verify vertex correspondence, calculate vertex displacements, and construct deformation cages where applicable.

Deformation algorithms. We will implement vertex-based interpolation as the main baseline and cage-based interpolation as an additional method. The cage method will use precomputed interpolation weights to transfer control-point movements to the enclosed mesh.

Interactive rendering. We will build an interface that displays the mesh and updates vertex positions as the user changes the stylization-strength parameter. Geometry updates will be separated from rendering to make the methods easier to compare and evaluate.

We plan to use existing mesh-processing and visualization libraries for file loading, rendering, and geometric analysis. The main implementation work will focus on deformation interpolation, cage-based manipulation, and quantitative evaluation.

We will initially work with precomputed stylized meshes, including compatible examples from existing methods where available. This avoids making diffusion-model reproduction a dependency for the core project.

6. Schedule

Week 1: November 13–19 — Setup and Baseline

- Review geometric stylization, mesh interpolation, and cage-based deformation.
- Set up the mesh processing and visualization environment.
- Collect or construct at least three original–stylized mesh pairs with matching topology.
- Implement vertex displacement interpolation.
- Produce initial visualizations at several stylization strengths.

Target: A working baseline that generates intermediate stylized meshes.

Week 2: November 20–26 — Cage Deformation and Milestone

- Implement a basic cage-based deformation pipeline.
- Integrate cage interpolation with the existing mesh viewer.
- Compare cage interpolation and direct vertex interpolation.
- Prepare preliminary visual results and numerical measurements.
- Submit the project milestone on November 22.

Target: Two working interpolation approaches and initial comparison results.

Week 3: November 27–December 3 — Interactive System and Evaluation

- Complete the interactive slider interface.
- Test deformation behavior across several objects and strength values.

- Implement geometric quality and performance measurements.
- Generate graphs and side-by-side comparison figures.
- Investigate geometry-aware interpolation if the baseline is stable.

Target: A functional interactive demo with quantitative results.

Week 4: December 4–8 — Final Results and Presentation

- Finalize experiments and visual comparisons.
- Record demonstration videos and capture high-quality renders.
- Prepare the final report and project webpage.
- Submit final deliverables by December 6.
- Present the project on December 7–8.

Target: A complete demo, reproducible results, a final report, and presentation materials.

7. Resources

Related Research

[1] Choi et al., "Image-Guided Geometric Stylization of 3D Meshes," CVPR 2026.

https://changwoonchoi.github.io/GeoStyle/

This work proposes image-guided geometric stylization using diffusion guidance and a coarse-to-fine deformation pipeline. It provides our primary motivation and a reference for cage-based geometric stylization. Our project focuses on continuous strength control over a resulting deformation rather than reproducing its entire generative pipeline.

[2] Dinh et al., "Geometry in Style: 3D Stylization via Surface Normal Deformation," CVPR 2025.

https://openaccess.thecvf.com/content/CVPR2025/html/Dinh_Geometry_in_Style_3D_Stylization_via_Surface_Normal_Deformation_CVPR_2025_paper.html

This method introduces surface-normal-based geometric deformation and differentiable ARAP regularization to achieve expressive stylization while preserving object identity. It motivates our investigation of geometric quality and possible regularization of intermediate deformations.

[3] Wynn et al., "Morpheus: Text-Driven 3D Gaussian Splat Shape and Color Stylization," CVPR 2025.

https://openaccess.thecvf.com/content/CVPR2025/papers/Wynn_Morpheus_Text-Driven_3D_Gaussian_Splat_Shape_and_Color_Stylization_CVPR_2025_paper.pdf

Morpheus introduces controls over appearance and shape stylization strength for 3D Gaussian Splatting. It provides relevant prior work on strength-controlled stylization, although our project focuses specifically on interpolation of explicit triangle meshes and analysis of intermediate mesh geometry.

Software and Hardware

We plan to use:

- **Python and NumPy:** Mesh deformation computations and analysis.
- **Trimesh and/or libigl:** Mesh processing, geometry calculations, and validation.
- **Polyscope or an equivalent interactive 3D viewer:** Visualization and interactive slider controls.
- **Blender:** Rendering final comparison images and demonstration videos.
- **PyTorch (optional):** Experimental geometry optimization or integration with existing stylization outputs.

Development and baseline interpolation will be performed on standard desktop hardware. A GPU may be used for advanced experiments, but we do not expect GPU-based diffusion-model training to be necessary for the core project.

We will refer to existing implementations and publicly available mesh assets where compatible, while implementing our own interpolation and evaluation components.

8. Expected Outcome

We expect to deliver an interactive geometric stylization system that lets users adjust shape exaggeration through a continuous control parameter.

The central contribution will be an implementation and evaluation of different geometric interpolation strategies, demonstrating how the choice of deformation representation affects controllability, visual smoothness, geometric validity, and runtime performance.

By combining interactive visualization with quantitative analysis, this project will explore how expressive stylized 3D geometry can become more directly controllable for graphics applications such as character design, digital modeling, and animation.

Our minimum successful outcome is a stable slider-based system with direct vertex interpolation, a working cage-based comparison, and documented quality and performance measurements. More advanced interpolation and perceptual control will be pursued if time permits.